Technical Note: Improvements in GEANT4 energy-loss model and the effect on low-energy electron transport in liquid water

I. Kyriakou[a)]
Medical Physics Laboratory, University of Ioannina Medical School, Ioannina 45110, Greece

S. Incerti
Centre d'Etudes Nucléaires de Bordeaux-Gradignan, CENBG, Chemin du Solarium, Université de Bordeaux, Gradignan 33175, France and Centre d'Etudes Nucléaires de Bordeaux-Gradignan, CENBG, Chemin du Solarium, CNRS/IN2P3, Gradignan 33175, France

Z. Francis
Department of Physics, Faculty of Sciences, Saint Joseph University, Mkalles, Beirut, Lebanon

(Received 27 February 2015; revised 22 April 2015; accepted for publication 12 May 2015; published 9 June 2015)

Purpose: The GEANT4-DNA physics models are upgraded by a more accurate set of electron cross sections for ionization and excitation in liquid water. The impact of the new developments on low-energy electron transport simulations by the GEANT4 Monte Carlo toolkit is examined for improving its performance in dosimetry applications at the subcellular and nanometer level.
Methods: The authors provide an algorithm for an improved implementation of the Emfietzoglou model dielectric response function of liquid water used in the GEANT4-DNA existing model. The algorithm redistributes the imaginary part of the dielectric function to ensure a physically motivated behavior at the binding energies, while retaining all the advantages of the original formulation, e.g., the analytic properties and the fulfillment of the f-sum-rule. In addition, refinements in the exchange and perturbation corrections to the Born approximation used in the GEANT4-DNA existing model are also made.
Results: The new ionization and excitation cross sections are significantly different from those of the GEANT4-DNA existing model. In particular, excitations are strongly enhanced relative to ionizations, resulting in higher W-values and less diffusive dose-point-kernels at sub-keV electron energies.
Conclusions: An improved energy-loss model for the excitation and ionization of liquid water by low-energy electrons has been implemented in GEANT4-DNA. The suspiciously low W-values and the unphysical long tail in the dose-point-kernel have been corrected owing to a different partitioning of the dielectric function.
[http://dx.doi.org/10.1118/1.4921613]

Key words: Monte Carlo, GEANT4, GEANT4-DNA, cross sections, dielectric function

1. INTRODUCTION

The main ingredient for a Monte Carlo track-structure (or event-by-event) simulation of charged-particle transport is the various interaction cross sections for the medium of interest.[1] For electrons, for example, these cross sections include elastic scattering, electronic excitation, and ionization.[2] In practice, there are two main difficulties in developing cross section models for track-structure simulations, namely, the need to cover the full slowing-down energy range (at least down to 10 eV for liquid water) which extends beyond the range of validity of the standard high-energy theories (like the Born and Bethe theories), and the need to include condensed-phase effects which play an increasingly important role at low energies. Apart from the above, there is also the more fundamental issue for all track-structure codes of whether (and how) quantum-mechanical uncertainties should be addressed since a low-energy electron certainly cannot be said to follow a well-defined trajectory in the liquid-water medium.[3,4] However, although trajectory simulation may not be strictly valid at electron energies below ~1 keV,[3] it has been shown to provide a reasonable approximation down to electron energies as low as some tens of eV, or even down to a few eV.[4] Thus, efforts in improving the physics models in track-structure Monte Carlo codes are expected to have an impact in emerging radiotherapeutic modalities like targeted radionuclide therapy and nanoparticle enhanced radiotherapy where dose calculations at the subcellular and nanometer level are important.[5,6]

The GEANT4-DNA cross section models have been fully described elsewhere.[7] Although various verification and validation results show that current GEANT4-DNA models are reasonably accurate for the transport of low-energy electrons (i.e., those with energy below 10 keV) in liquid water, several deficiencies have been recognized:

- The electron excitation cross sections in GEANT4-DNA compare poorly to other models available in the literature and to experimental measurements.[7]
- The average energy per ion pair (W-value) for electron sources simulated with GEANT4-DNA is suspiciously

lower compared to other Monte Carlo simulations and to experimental measurements.[8]
- The dose-point-kernel (DPK) of low-energy electrons exhibits an unrealistic long tail extending to large distances.

In this work, we present an algorithm for an improved implementation of a model dielectric response function that ensures a physically motivated behavior at the binding energies and the fulfillment of the f-sum-rule. The algorithm is applied into the dielectric function that is currently used in GEANT4-DNA and, along with refinements made in the existing Born corrections, improves the performance of GEANT4 for low-energy electrons in liquid water.

2. MATERIALS AND METHODS

2.A. Dielectric response function

Inelastic collisions of electrons with atoms leading to ionization and (electronic) excitations represent the main energy-loss mechanism in the energy range of interest in the present work. In the nonrelativistic plane-wave first Born approximation (hereafter simply Born approximation) for condensed media, the double-differential cross section in energy transfer E and momentum transfer q per atom or molecule takes the form[9]

$$\frac{d^2\sigma_{\rm Born}(E,q;T)}{dE\,dq} = \frac{1}{\pi a_0 N\,T\,q}\,{\rm Im}\left[-\frac{1}{\varepsilon(E,q)}\right], \tag{1}$$

where a_0 is the Bohr radius, N is the density of atoms or molecules, T is the electron kinetic energy, $\varepsilon(E,q)$ is the dielectric function of the material, and Im[...] denotes the imaginary part of the argument. For amorphous materials, like liquid water, it is generally justified to treat q as being a scalar quantity. From Eq. (1), we can determine by integrations the energy loss and momentum transfer (or scattering angle) in an inelastic collision as well as the inelastic mean free path, which are the main ingredients of a track-structure code. Thus, the only nontrivial parameter for calculating ionization and excitation cross sections (in the Born approximation) for liquid water is the dielectric function, $\varepsilon(E,q)$, which accounts for the screening and polarization properties of the condensed phase.

It is recognized[10,11] that there are three major models of the dielectric function of liquid water developed and refined through a series of studies: the Ritchie *et al.* model,[12] the Dingfelder *et al.* model,[13–15] and the Emfietzoglou *et al.* model.[16–20] All these models are based on an analytic parameterization of the experimental data (those of Heller *et al.*[21] or Hayashi *et al.*[22]) for the dielectric function of liquid water at the optical limit ($q \approx 0$) using a linear superposition of Drude-like functions, and a set of dispersion relations to extend the dielectric function to $q \neq 0$. They differ in the details of the parameterization, the behavior of the model at energies close to the ionization thresholds (binding energies), and the dispersion relations adopted. Despite its shortcomings,[23] a Drude-like description of the dielectric function has been extensively used in track-structure codes for liquid water (OREC/NOREC,[24] PARTRAC,[25] PITS,[26] KURBUC,[27] among others[28–31]). Recently, the Dingfelder *et al.* model[13] (with modifications[31]) has been used in the general-purpose code PENELOPE for extending its applicability to the simulation of full slowing-down electron tracks in liquid water.[32] The particularly convenient analytic properties of the Drude function, not shared by other more elaborate models,[23] are the main reason for its widespread application.[9,15]

In GEANT4-DNA existing model,[7] the calculation of excitation and ionization cross sections is based on the Emfietzoglou *et al.* model[16–18] of the dielectric function with Heller's optical data[21] (except for the ionization cross section of the K-shell which is based on the binary-encounter-with-exchange, BEAX, model). The first step is to express the so-called energy-loss function, ${\rm Im}[-1/(\varepsilon(E,q))]$, that enters in Eq. (1), in terms of the real and imaginary parts of the dielectric function

$${\rm Im}\left[-\frac{1}{\varepsilon(E,q)}\right] = \frac{{\rm Im}[\varepsilon(E,q)]}{{\rm Re}[\varepsilon(E,q)]^2+{\rm Im}[\varepsilon(E,q)]^2}, \tag{2}$$

where the numerator, ${\rm Im}[\varepsilon(E,q)]$, describes the absorption properties of the medium and the denominator, ${\rm Re}[\varepsilon(E,q)]^2 + {\rm Im}[\varepsilon(E,q)]^2 = |\varepsilon(E,q)|^2$, the screening properties. From the correspondence between the imaginary part of the dielectric function and the oscillator strength,[9] we can write the following partitioning expression:

$$\begin{aligned}{\rm Im}[\varepsilon(E,q=0)] = &\sum_n^{\rm ioniz.} {\rm Im}[\varepsilon_n(E,q=0)]\\ &+\sum_k^{\rm excit.} {\rm Im}[\varepsilon_k(E,q=0)],\end{aligned} \tag{3}$$

where the index n runs over the various ionization shells and the index k runs over the discrete excitation levels. Equation (3) is then fitted to the optical data of Heller *et al.*[21] using Drude-like functions to represent the ${\rm Im}[\varepsilon_{n/k}(E,q=0)]$ terms. To ensure the self-consistency of the model, the fitting procedure is carried out under the constraint imposed by the so-called f-sum-rule,

$$\int_0^\infty E\,{\rm Im}[\varepsilon(E,q=0)]dE = \frac{\pi}{2}E_{\rm pl}^2, \tag{4}$$

where $E_{\rm pl}$ is the (nominal) plasmon energy. Due to the particular form of the Drude functions, the real part of the dielectric function can be also expressed analytically through the Kramers–Kronig relation. Finally, the extension of the dielectric function to $q \neq 0$ is carried out by means of suitable dispersion relations for the Drude coefficients.

The deconvolution of the optical data using Eq. (3) to obtain the model parameters related to the representation of each ${\rm Im}[\varepsilon_{n/k}(E,q=0)]$ term, although based on some physically motivated general constraints (e.g., sum rules and threshold energies), is empirical and largely heuristic. This problem is common to all empirical optical-data models of the dielectric function. Moreover, for any damping-inclusive model (such as the Drude model), the model parameters obtained by the deconvolution of the optical data are not uniquely defined.[33]

In general, only *ab initio* calculations of optical data have the potential to resolve the above problems.[34,35]

2.B. Truncation algorithm

A problem with the above scheme is that it allows for a nonvanishing contribution of $\mathrm{Im}[\varepsilon_n(E,q)]$ below the binding energies (B_n), which is clearly unphysical.[13] A common strategy to overcome the above problem, also adopted in the GEANT4-DNA existing model, is to "cut" the Drude functions at B_n, so that they vanish below the ionization thresholds. However, if the truncation of the Drude functions takes place after the fitting (which is more convenient), it will naturally lead to a smaller oscillator strength. As a result, without proper precautions, truncated Drude functions may violate (or worsen) the f-sum-rule. Moreover, if the imaginary part of the dielectric function is expressed as a truncated Drude function, the mathematical expressions for the real part of the dielectric function, obtained through the Kramers–Kronig relation, are significantly more complicated.[13] This problem is general and not limited to the Drude function. For example, the Kramers–Kronig pair of the Lindhard function with an energy-gap[36] is more complicated mathematically, and one expects the calculations to become intractable with truncated or gap-inclusive Mermin functions.

To overcome these problems, we have implemented a truncation algorithm (see the Appendix) which aims to account for the vanishing contribution of $\mathrm{Im}[\varepsilon_n(E,q)]$ at $E < B_n$ and also to provide an exponential smoothing around B_n.[18] The algorithm essentially redistributes the oscillator strength, i.e., $\mathrm{Im}[\varepsilon(E,q=0)]$, among the various ionization shells (n) and excitation levels (k) in a sum-rule conserving manner, ensuring that

$$\mathrm{Im}[\varepsilon(E,q=0)] = \sum_{n/k}\mathrm{Im}[\varepsilon_{n/k}(E,q=0)] = \sum_{n/k}\mathrm{Im}[\varepsilon^{(\mathrm{mod})}_{n/k}(E,q=0)], \tag{5}$$

where $\mathrm{Im}[\varepsilon_{n/k}(E,q=0)]$ is computed from the original (untruncated) Drude functions and $\mathrm{Im}[\varepsilon^{(\mathrm{mod})}_{n/k}(E,q=0)]$ from the modified (truncated) Drude functions (see the Appendix). Thus, we obtain a more realistic behavior at the binding energies while retaining the shape and magnitude of $\mathrm{Im}[\varepsilon(E,q=0)]$ deduced from the fit to the optical data [through Eq. (3)]. The practical importance of Eq. (5) (i.e., the conservation of $\mathrm{Im}[\varepsilon(E,q=0)]$) is that the real part of the dielectric function, $Re[\varepsilon(E,q=0)]$, remains unchanged regardless of whether the original Drude functions ($\mathrm{Im}[\varepsilon_{n/k}(E,q=0)]$) or the modified Drude functions ($\mathrm{Im}[\varepsilon^{(\mathrm{mod})}_{n/k}(E,q=0)]$) are used.

The extension of the dielectric function to $q \neq 0$ is carried out in the same way with the original formulation of the model. However, due to the mixing of states, the extension of the $\mathrm{Im}[\varepsilon_{n/k}(E,q=0)]$ of a particular state (n or k) to $q \neq 0$ has now some (small) contributions from other states which may have different q-dependencies. In practice, this "mixing-of-state" effect is negligible because the "extra" contributions of oscillator strength from other states are always (very) small compared to the oscillator strength of any given state.

It should be noted that the algorithm is essentially independent of the model used to represent the dielectric function (see the Appendix). Its practical significance is that it allows the inclusion of binding energies in a sum-rule conserving manner while retaining the convenience of working with the Kramers–Kronig pairs of the untruncated dielectric function.

2.C. Born corrections

Low-energy corrections to the Born approximation are of two kinds, namely, corrections for exchange and correlation in electron–electron interactions and corrections for the departure from the plane-wave first-order perturbation theory.[37–40] In the present work, we express the corrected (inelastic) cross section by a series of terms as follows:

$$\sigma^{\mathrm{corrected}}_{\mathrm{total}}(T) = \sigma^{\mathrm{direct}}_{\mathrm{ioniz}}(T') + \sigma^{\mathrm{exchange}}_{\mathrm{ioniz}}(T') - \sigma^{\mathrm{interference}}_{\mathrm{ioniz}}(T') + \sigma_{\mathrm{excit}}(T'), \tag{6}$$

where the direct term is the (uncorrected) Born cross section as obtained from Eq. (1), the exchange and interference terms follow from the Mott-like correction, and the kinematic transformation $T \to T'$, with $T' = T + 2E_k$ in the case of excitation and $T' = T + B_n + U_n$ in the case of ionization (U_n being the average kinetic energy of the electron at the nth shell), follows from the Coulomb correction (for more details, see Ref. 18). Equation (6) introduces two important modifications to the GEANT4-DNA existing model[41] that are important at low energies. First, we do not use the factor $[1-(B_n/T)^{1/2}]$ to multiply the interference term. This factor, which approaches one at high electron energies ($T >> B_n$), leads to a vanishing interference term at low electron energies ($T \to B_n$). The latter leads to an unphysical increase of the exchange-corrected ionization cross section above the value of the Born approximation.[18] Second, the kinematic transformation ($T \to T'$) is applied here to all terms on the right-hand-side of Eq. (6) since it is meant to account for effects beyond first order.

3. RESULTS AND DISCUSSION

3.A. Effect on cross sections

The effect of the truncation algorithm is shown in Fig. 1 which depicts the imaginary part of the model dielectric function for the excitation [panel (a)] and ionization [panel (b)] states of liquid water.

The partitioning of $\mathrm{Im}[\varepsilon(E,q=0)]$ is based on the fitting and deconvolution of the optical data of Heller *et al.*[21] using Drude-like functions as explained elsewhere.[18] Comparison is made between the results of the present model which uses the truncation algorithm (see the Appendix) and the results based on the GEANT4-DNA existing model. The effect of the algorithm is to enhance the contribution of the excitation states [see panel (a)] while eliminating the contribution of

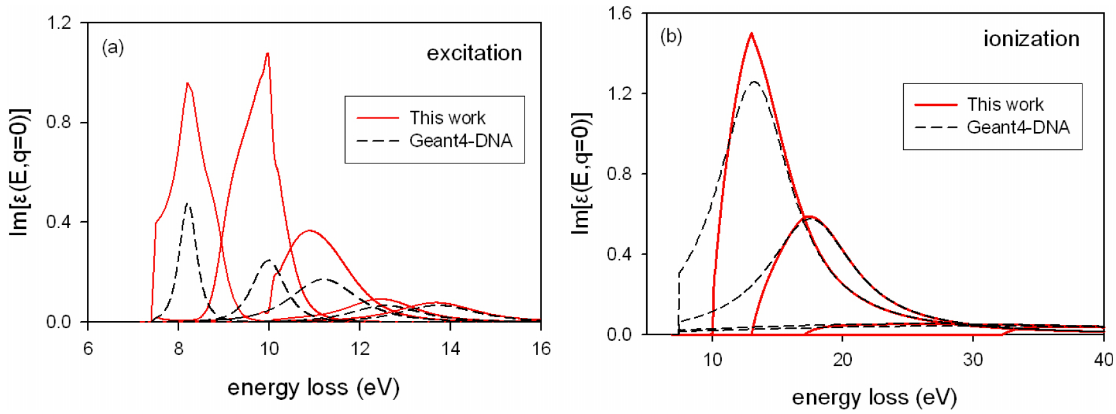

FIG. 1. The imaginary part of the Emfietzoglou model dielectric function for the different excitation states [panel (a)] and ionization shells [panel (b)] of liquid water which is based on the parameterization of the optical data of Heller *et al.* (Ref. 21). Full-line: results of the present model using the truncation algorithm; Dashed-line: results based on the GEANT4-DNA existing model.

each ionization state below the corresponding binding energy (B_n) with a concomitant smoothing at the near-threshold region [see panel (b)].

In Fig. 2, we show the fractional contribution of ionization and excitation to the total inelastic cross section as a function of the electron energy from 10 keV down to 10 eV. The present model results in much higher (lower) excitation (ionization) yield compared to the GEANT4-DNA existing model. This is a direct consequence of the effect of the truncation algorithm implemented in the present dielectric function which redistributes part of the ionization oscillator strength to excitations (see Fig. 1). Although the difference between the present and GEANT4-DNA models is noticeable over the entire energy range, it increases rapidly below 100 eV where the ionization channels are gradually turned off because of the condition $\mathrm{Im}[\varepsilon_n(E < B_n)] = 0$ (that holds for any q). An additional reason for the lower ionization contribution in the present model is the somewhat different corrections to the Born approximation adopted. The factor $[1-(B_n/T)^{1/2}]$ used in the GEANT4-DNA existing model reduces the magnitude of the interference term which is subtracted from the Born cross section, thus, resulting in larger ionization cross sections.

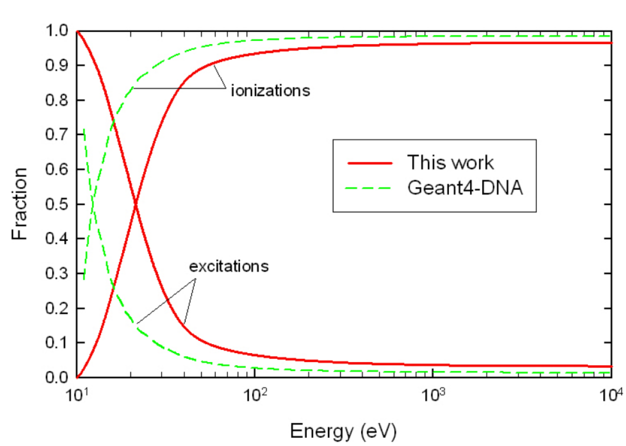

FIG. 2. The contribution of ionizations and excitations to the total inelastic cross section as a function of electron energy. Full-line: results of the present model; broken line: results based on the GEANT4-DNA existing model (Refs. 7 and 41).

3.B. Effect on simulations

For all simulation results, electrons (either primary or secondary) are followed down to 8 eV. An important benchmark for the excitation and ionization cross sections is the W-value which is the average energy expended per ion pair formed during the complete slowing-down of an initial particle of a given energy. For electron sources, the W-value is independent of the elastic scattering model. It has been recently noted[8] that GEANT4-DNA predicts much lower W-values for liquid water compared to other similar track-structure codes. In Fig. 3, we present the energy-dependence of the W-value as obtained by GEANT4-DNA using both the existing (default) and the present model. The results

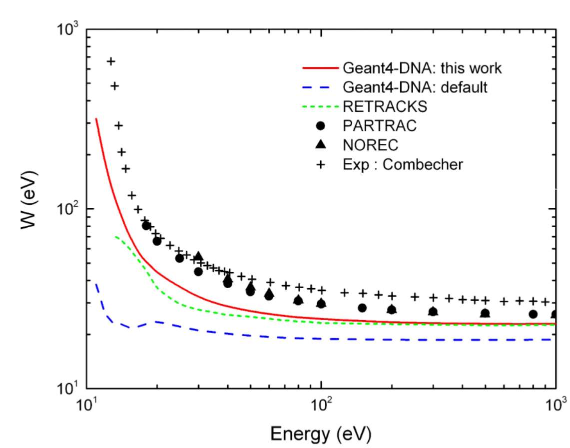

FIG. 3. The energy-dependence of the W-value of water. Full-line: simulation results with GEANT4-DNA using the present model; long dashed line: simulation results with GEANT4-DNA using the existing (default) model; short dashed line: simulation results of the liquid water track-structure code RETRACKS (Ref. 42); circles: simulation results of the liquid water track-structure code PARTRAC (Ref. 25); triangles: simulation results of the liquid water track-structure code NOREC (Ref. 24); crosses: experimental data on gaseous water (Ref. 43).

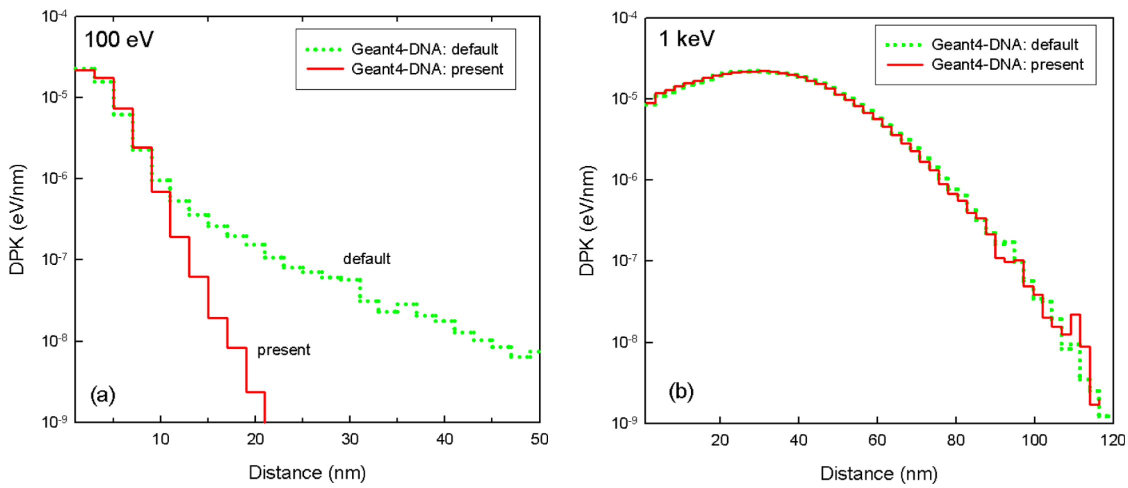

FIG. 4. The DPK in liquid water for monoenergetic electron sources with initial energy of 100 eV [panel (a)] and 1 keV [panel (b)]. Full-line: simulation results with GEANT4-DNA using the present model; dotted line: simulation results with GEANT4-DNA using the existing (default) model.

are compared against the liquid-water track-structure codes NOREC,[24] PARTRAC,[25] and RETRACKS.[42] Also shown are the experimental data of Combecher[43] for gaseous water.

The present model brings much better agreement with the other track-structure codes. The W-value sensitively depends on the relative magnitude of the excitation and ionization cross sections. The larger the excitation-to-ionization cross section ratio is, the higher the W-value since a smaller number of ion pairs will be formed (for the same electron energy dissipated). This expectation is confirmed here; the higher W-values of the present simulations follow (mainly) from the enhanced excitation contribution in the present dielectric model (Fig. 1) and inelastic cross section (Fig. 2). Some difference with the experimental data for gaseous water is expected and confirms the well-established higher ionization yield of the liquid phase compared to the gas phase.[44] In essence, the W-value of the gas phase represents an upper bound for the W-value of the liquid phase.

DPKs, defined as the average energy deposited per unit distance along the radius of a sphere centered at the starting point of the particle track, provide useful information on the spatial distribution of energy deposition. DPKs for monoenergetic electron sources with initial energy of 100 eV and 1 keV simulated by GEANT4-DNA are presented in Fig. 4 using the existing (default) and the present model (with respect to elastic scattering, the GEANT4-DNA default partial-wave model has been used in all the simulations).

It can be seen that, although for 1 keV electrons both DPKs are in good agreement, there is significant discrepancy for the 100 eV electrons. Specifically, the default model predicts a DPK with a long tail indicative of very diffuse electron tracks. The long tail disappears when the default model is replaced by the present model. At large distances from the track, most electrons have very low energies and lose energy predominantly to excitations. The very small excitation cross section of the default model allows these very low-energy electrons to diffuse much longer distances in the medium before their energy falls below the transport cutoff (~8 eV). The effect is not noticeable at high energies (above ~1 keV) because the electrons are energetic enough to reach relatively much larger penetration distances.

4. CONCLUSION

An improved energy-loss model for the ionization and excitation of liquid water by low-energy electrons has been implemented in GEANT4-DNA and shown to enhance the performance of the GEANT4 Monte Carlo toolkit with respect to the simulation of W-values and dose-point-kernels at sub-keV electron energies. The improvements are brought about by the implementation of a truncation algorithm in the Emfietzoglou model for the dielectric response function of liquid water and refinements made on the low-energy corrections to the Born approximation.

ACKNOWLEDGMENTS

The authors are indebted to Dimitris Emfietzoglou for valuable discussions and comments on the paper. This work was partly funded by the European Space Agency under Contract No. BioRad II AO7146-4000107387/12/NL/AK (2013–2016). The authors are members of the GEANT4-DNA collaboration.

APPENDIX: TRUNCATION ALGORITHM

Hereafter, for clarity, we write $\mathrm{Im}[\varepsilon_{n/k}(E,q=0)]$ as $\mathrm{Im}[\varepsilon_{n/k}(E)]$. The algorithm modifies the $\mathrm{Im}[\varepsilon_n(E)]$ terms in such a way that they are truncated at the binding energies, B_n, and smoothed around B_n. To conserve the original f-sum, the truncated part of $\mathrm{Im}[\varepsilon_n(E)]$, as long as it is above the minimum binding energy (i.e., $E > B_{n=1}$), is partitioned to lower ionization shells (i.e., those having smaller B_n). This partitioning is weighted according to the relative magnitude of each $\mathrm{Im}[\varepsilon_n(E)]$. If the truncated part of $\mathrm{Im}[\varepsilon_n(E)]$ falls below the minimum binding energy (i.e., $E \leq B_{n=1}$), it is partitioned to the excitation states, i.e., to $\mathrm{Im}[\varepsilon_k(E)]$, with resonance energy below the ionization threshold (i.e., $E_k \le B_{n=1}$). This partitioning is weighted according to the relative magnitude of each $\mathrm{Im}[\varepsilon_k(E)]$. The mathematical implementation of the truncation algorithm follows.

- For ionizations,

$$\mathrm{Im}[\varepsilon_n^{(\mathrm{mod})}(E)] = \{\mathrm{Im}[\varepsilon_n(E)] + S_n + S_{>n} + C_{>n}\}\, C_n, \quad \text{(A1)}$$

where

$$C_n = \mathrm{H}(E - B_n), \quad \text{(A2)}$$

$$C_{>n} = \sum_{j=n+1}^{n_{\max}} \mathrm{Im}[\varepsilon_j(E)]\,\Theta(B_j - E) \times \left(\frac{\mathrm{Im}[\varepsilon_n(E)]}{\sum_{i=1}^{j-1}\mathrm{Im}[\varepsilon_i(E)]\,\mathrm{H}(E-B_i)}\right), \quad \text{(A3)}$$

$$S_{>n} = \sum_{j=n+1}^{n_{\max}} \mathrm{Im}[\varepsilon_j(B_j)]\exp(B_j - E)\,\mathrm{H}(E-B_j) \times \left(\frac{\mathrm{Im}[\varepsilon_n(E)]}{\sum_{i=1}^{j-1}\mathrm{Im}[\varepsilon_i(E)]}\right), \quad \text{(A4)}$$

$$S_n = -\mathrm{Im}[\varepsilon_n(B_n)]\exp(B_n - E). \quad \text{(A5)}$$

- For excitations,

$$\mathrm{Im}[\varepsilon_k^{(\mathrm{mod})}(E)] = \{\mathrm{Im}[\varepsilon_k(E)] + S_{n=1} + C_n\}\, C_k, \quad \text{(A6)}$$

where

$$C_k = \Theta(E - B_{\mathrm{gap}}), \quad \text{(A7)}$$

$$C_n = \sum_{n=1}^{n_{\max}} \mathrm{Im}[\varepsilon_n(E)]\,\Theta(B_1 - E) \times \left(\frac{\mathrm{Im}[\varepsilon_k(E)]\,\Theta(B_1 - E_k)}{\sum_{j=1}^{k_{\max}}\mathrm{Im}[\varepsilon_j(E)]\,\Theta(B_1 - E_j)}\right), \quad \text{(A8)}$$

$$S_{n=1} = \mathrm{Im}[\varepsilon_{n=1}(B_1)]\exp(B_1 - E)\, H(E - B_1) \times \left(\frac{\mathrm{Im}[\varepsilon_k(E)]}{\sum_{j=1}^{k_{\max}}\mathrm{Im}[\varepsilon_j(E)]}\right), \quad \text{(A9)}$$

where $\Theta(\ldots)$ and $H(\ldots)$ are unit step functions with $\Theta(0) = 1$ and $H(0) = 0$.

[a]Electronic mail: ikyriak@cc.uoi.gr

[1]H. Nikjoo, S. Uehara, D. Emfietzoglou, and F. A. Cucinotta, "Monte Carlo track structure codes in radiation research–A review," Radiat. Meas. **41**, 1052–1074 (2006).

[2]S. Uehara, H. Nikjoo, and D. T. Goodhead, "Comparison and assessment of electron cross sections for Monte Carlo track structure codes," Radiat. Res. **152**, 202–213 (1999).

[3]R. M. Thomson and I. Kawrakow, "On the Monte Carlo simulation of electron transport in the sub-1 keV energy range," Med. Phys. **38**, 4531–4534 (2011).

[4]D. Liljequist and H. Nikjoo, "On the validity of trajectory methods for calculating the transport of very low energy (<1 keV) electrons in liquids and amorphous media," Radiat. Phys. Chem. **99**, 45–52 (2014).

[5]D. Sarrut *et al.*, "A review of the use and potential of the GATE Monte Carlo simulation code for radiation therapy and dosimetry applications," Med. Phys. **41**, 064301 (14pp.) (2014).

[6]M. K. K. Leung, J. C. L. Chow, B. D. Chithrani, M. J. G. Lee, B. Oms, and D. A. Jaffray, "Irradiation of gold nanoparticles by x-rays: Monte Carlo simulation of dose enhancements and the spatial properties of the secondary electrons production," Med. Phys. **38**, 624–631 (2011).

[7]S. Incerti *et al.*, "Comparison of GEANT4 very low energy cross section models with experimental data in water," Med. Phys. **37**, 4692–4708 (2010).

[8]O. N. Vassiliev, "Electron slowing-down spectra in water for electron and photon sources calculated with the Geant4-DNA code," Phys. Med. Biol. **57**, 1087–1094 (2012).

[9]H. Nikjoo, S. Uehara, and D. Emfietzoglou, *Interaction of Radiation with Matter* (CRC Press Taylor & Francis, Boca Raton, 2012).

[10]M. Dingfelder, "Track structure: Time evolution from physics to chemistry," Radiat. Prot. Dosim. **122**, 16–21 (2006).

[11]M. Dingfelder, "Track-structure simulations for charged particles," Health Phys. **103**, 590–595 (2012).

[12]R. H. Ritchie, R. N. Hamm, J. E. Turner, H. A. Wright, and W. E. Bolch, "Radiation interactions and energy transport in the condensed phase," in *Physical and Chemical Mechanisms in Molecular Radiation Biology*, edited by W. A. Glass and M. N. Varma (Plenum Press, New York, 1991), pp. 99–136.

[13]M. Dingfelder, D. Hantke, M. Inokuti, and H. G. Paretzke, "Electron inelastic-scattering cross sections in liquid water," Radiat. Phys. Chem. **53**, 1–18 (1998).

[14]M. Dingfelder and M. Inokuti, "The Bethe surface of liquid water," Radiat. Environ. Biophys. **38**, 93–96 (1999).

[15]M. Dingfelder, "Updated model for dielectric response function of liquid water," Appl. Radiat. Isot. **83**, 142–147 (2014).

[16]D. Emfietzoglou and M. Moscovitch, "Inelastic collision characteristics of electrons in liquid water," Nucl. Instrum. Methods Phys. Res., Sect. B **193**, 71–78 (2002).

[17]D. Emfietzoglou, "Inelastic cross-sections for electron transport in liquid water: A comparison of dielectric models," Radiat. Phys. Chem. **66**, 373–385 (2003).

[18]D. Emfietzoglou and H. Nikjoo, "The effect of model approximations on single-collision distributions of low-energy electrons in liquid water," Radiat. Res. **163**, 98–111 (2005).

[19]D. Emfietzoglou, F. Cucinotta, and H. Nikjoo, "A complete dielectric response model for liquid water: A solution of the Bethe ridge problem," Radiat. Res. **164**, 202–211 (2005).

[20]D. Emfietzoglou and H. Nikjoo, "Accurate electron inelastic cross sections and stopping powers for liquid water over the 0.1-10 keV range based on an improved dielectric description of the Bethe surface," Radiat. Res. **167**, 110–120 (2007).

[21]J. M. Heller, R. N. Hamm, R. D. Birkhoff, and L. R. Painter, "Collective oscillation in liquid water," J. Chem. Phys. **60**, 3483–3486 (1974).

[22]H. Hayashi, N. Watanabe, Y. Udagawa, and C. C. Kao, "The complete optical spectrum of liquid water measured by inelastic x-ray scattering," Proc. Natl. Acad. Sci. U. S. A. **97**, 6264–6266 (2000).

[23]D. Emfietzoglou, I. Kyriakou, I. Abril, R. Garcia-Molina, and H. Nikjoo, "Inelastic scattering of low energy electrons in liquid water computed from optical-data models of the Bethe surface," Int. J. Radiat. Biol. **88**, 22–28 (2012).

[24]V. A. Semenenko, J. E. Turner, and T. B. Borak, "NOREC a Monte Carlo code for simulating electron tracks in liquid water," Radiat. Environ. Biophys. **42**, 213–217 (2003).

[25]M. Dingfelder, R. H. Ritchie, J. E. Turner, W. Friedland, H. G. Paretzke, and R. N. Hamm, "Comparisons of calculations with PARTRAC and NOREC: Transport of electrons in liquid water," Radiat. Res. **169**, 584–594 (2008).

[26]W. E. Wilson, J. H. Miller, D. J. Lynch, R. R. Lewis, and M. Batdorf, "Analysis of low-energy electron track structure in liquid water," Radiat. Res. **161**, 591–596 (2004).

[27]T. Liamsuwan, D. Emfietzoglou, S. Uehara, and H. Nikjoo, "Microdosimetry of low-energy electrons," Int. J. Radiat. Biol. **88**, 899–907 (2012).

[28]H. Tomita, M. Kai, T. Kusama, and A. Ito, "Monte Carlo simulation of physicochemical processes of liquid water radiolysis," Radiat. Environ. Biophys. **36**, 105–116 (1997).

[29]D. Emfietzoglou, K. Karava, G. Papamichael, and M. Moscovitch, "Monte Carlo simulation of the energy loss of low-energy electrons in liquid water," Phys. Med. Biol. **48**, 2355–2371 (2003).

[30]C. J. Tung, T. C. Chao, H. W. Hsieh, and W. T. Chan, "Inelastic interactions of low-energy electrons with biological media," Nucl. Instrum. Methods Phys. Res., Sect. B **262**, 231–239 (2007).

[31]K. Wiklund, J. M. Fernández-Varea, and B. K. Lind, " A Monte Carlo program for the analysis of low-energy electron tracks in liquid water," Phys. Med. Biol. **56**, 1985–2003 (2011).

[32]J. M. Fernández-Varea *et al.*, "Limitations (and merits) of PENELOPE as a track-structure code," Int. J. Radiat. Biol. **88**, 66–70 (2012).

[33]C. T. Chantler and J. D. Bourke, "Momentum-dependent lifetime broadening of electron energy loss spectra: Sum rule constraints and an asymmetric rectangle toy model," Phys. Rev. B **90**, 174306 (2014).
[34]A. P. Sorini, J. J. Kas, J. J. Rehr, M. P. Prange, and Z. H. Levine, "An initio calculations of electron inelastic mean free paths and stopping powers," Phys. Rev. B **74**, 165111 (2006).
[35]C. T. Chantler and J. D. Bourke, "Electron inelastic mean free path theory and density functional theory resolving discrepancies for low-energy electrons in copper," J. Phys. Chem. A **118**, 909–914 (2014).
[36]Z. H. Levine and S. G. Louie, "New model dielectric function and exchange–correlation potential for semiconductors and insulators," Phys. Rev. B **25**, 6310–6315 (1982).
[37]D. Emfietzoglou, I. Kyriakou, I. Abril, R. Garcia-Molina, and H. Nikjoo, "Inelastic cross sections for low-energy electrons in liquid water: Exchange and correlation effects," Radiat. Res. **180**, 499–513 (2013).
[38]D. Emfietzoglou *et al.*, "Electron inelastic mean free paths in biological matter based on dielectric theory and local-field corrections," Nucl. Instrum. Methods Phys. Res., Sect. B **267**, 45–52 (2009).
[39]D. Emfietzoglou, I. Kyriakou, I. Abril, and R. Garcia-Molina, "The effect of static many-body local field corrections to inelastic electron scattering in condensed media," J. Appl. Phys. **114**, 144907 (10pp.) (2013).
[40]J. M. Fernández-Varea, R. Mayol, D. Liljequist, and F. Salvat, "Inelastic scattering of electrons in solids from a generalized oscillator strength model using optical and photoelectric data," J. Phys.: Condens. Matter **5**, 3593–3610 (1993).
[41]C. Villagrasa, Z. Francis, and S. Incerti, "Physics models implemented in the Geant4-DNA extension of the Geant4 toolkit for calculating initial radiation damage at the molecular level," Radiat. Prot. Dosim. **143**, 214–218 (2011).
[42]I. Plante and F. A. Cucinotta, "Cross sections for the interactions of 1 eV-100 MeV electrons in liquid water and application to Monte-Carlo simulation of HZE radiation tracks," New J. Phys. **11**, 063047 (24pp.) (2009).
[43]D. Combecher, "Measurement of W values of low-energy electrons in several gases," Radiat. Res. **84**, 189–218 (1980).
[44]J. E. Turner, H. G. Paretzke, R. N. Hamm, H. A. Wright, and R. H. Ricthie, "Comparative study of electron energy deposition and yields in water in the liquid and vapor phases," Radiat. Res. **92**, 47–60 (1982).